

Methods of factorising

Four methods in a fixed order — common factor, formula, grouping, and the trinomial split.

LESSON 106–107

2 × 40 MIN

MATEMATIKA 7, §48

S8 2.4

LESSON CLOCK

10'

25'

25'

20'

5'

The order of the methods	10 min
Splitting the middle term	25 min
Combining methods	25 min
When to stop	20 min
Homework	5 min

LEARNING OBJECTIVES

- Apply the four methods of factorising in the right order.
- Split the middle term of a quadratic trinomial into two.
- Combine methods where one alone is not enough.
- Recognise when a polynomial is already fully factorised.

TEXTBOOK

National	pp. 311–318
Cambridge	Stage 8, pp. 25–30
Workbook	Exercise 2.4

01

Explanation

The order of the methods

Always work down this list. Taking the steps out of order is what makes a factorisation look impossible.

STEP	METHOD	LOOK FOR
1	common factor	something in every term
2	an abridged formula	two terms, or a perfect square
3	grouping	four terms
4	splitting the middle term	a quadratic trinomial

STEP 1 IS NEVER SKIPPED

$2x^2 - 18$ looks like nothing until the 2 comes out; $3x^2 + 12x + 12$ becomes $3(x + 2)^2$ the same way. Take the common factor first, every single time.

Splitting the middle term

For $x^2 + px + q$, find two numbers whose **product** is q and whose **sum** is p . Split the middle term into those two and group.

TRINOMIAL	PRODUCT	SUM	THE TWO NUMBERS	FACTORISED
$x^2 + 7x + 12$	12	7	3, 4	$(x + 3)(x + 4)$
$x^2 - 5x + 6$	6	-5	-2, -3	$(x - 2)(x - 3)$
$x^2 + x - 12$	-12	1	4, -3	$(x + 4)(x - 3)$
$x^2 - 2x - 15$	-15	-2	-5, 3	$(x - 5)(x + 3)$

THE SIGN OF THE PRODUCT COMES FIRST

A negative product means the two numbers have opposite signs; a positive product means they match, and then the sign of the middle term tells you which sign both take. Deciding the signs before hunting for the numbers halves the work.

Combining methods

EXPRESSION	STEPS	RESULT
$2x^2 - 18$	common factor, then difference of squares	$2(x - 3)(x + 3)$
$3x^2 + 12x + 12$	common factor, then a perfect square	$3(x + 2)^2$
$x^3 - x$	common factor, then difference of squares	$x(x - 1)(x + 1)$
$2x^2 + 10x + 12$	common factor, then split the middle term	$2(x + 2)(x + 3)$
$a^3 + a^2 - a - 1$	grouping, then difference of squares	$(a - 1)(a + 1)^2$

ONE METHOD IS RARELY THE WHOLE ANSWER

Stopping at $2(x^2 - 9)$ loses the marks that the second step carries. Ask after every step whether any bracket can still be broken down.

When to stop

EXPRESSION	FULLY FACTORISED?	WHY
$2(x - 3)(x + 3)$	yes	each bracket is linear
$2(x^2 - 9)$	no	$x^2 - 9$
$(x + 1)(x^2 + 1)$	yes	a sum of squares does not factorise
$(x - 2)(x^2 + 2x + 4)$	yes	the incomplete square does not factorise
$x^2 + x + 1$	yes — it is prime	no two whole numbers give the product and the sum

CHECK BY EXPANDING, ALWAYS

A factorisation is finished when no bracket can be broken further, and it is correct when expanding it returns the original polynomial exactly. Both checks take seconds.

02 Worked examples

EXAMPLE 1 Factorise $x^2 + x - 12$.

- ① Product -12 , sum 1 .
A negative product: opposite signs.
- ② The numbers are 4 and -3 .
- ③ $x^2 + 4x - 3x - 12$
- ④ $x(x + 4) - 3(x + 4) = (x + 4)(x - 3)$

ANSWER

$$(x + 4)(x - 3)$$

EXAMPLE 2 Factorise $2x^2 + 10x + 12$ completely.

- 1 Common factor 2: $2(x^2 + 5x + 6)$.
- 2 Product 6, sum 5: the numbers are 2 and 3.
- 3 $= 2(x + 2)(x + 3)$
- 4 Check by expanding ✓

ANSWER

$$2(x + 2)(x + 3)$$

EXAMPLE 3 Factorise $a^3 + a^2 - a - 1$.

- 1 Group: $a^2(a + 1) - (a + 1)$.
- 2 $= (a + 1)(a^2 - 1)$
- 3 $a^2 - 1 = (a - 1)(a + 1)$
Do not stop here.
- 4 $= (a - 1)(a + 1)^2$

ANSWER

$$(a - 1)(a + 1)^2$$

03 Interactive model

Put five mixed expressions on the board and ask only which method each needs, without factorising; choosing the method is the skill, and it can be practised on its own.

Which method does this one need? $6x^2 + 9x$ needs: $x^2 - 25$ needs: $ax + ay + bx + by$ needs:

$x^2 + 7x + 12$ needs:

a common factor

a formula

grouping

splitting

$2x^2 - 18$ needs:

a common factor only

a formula only

a common factor then a formula

grouping

For $x^2 - 5x + 6$ the two numbers are:

2, 3

-2, -3

-1, -6

1, 6

A negative product means the numbers:

are both positive

are both negative

have opposite signs

are equal

$2(x^2 - 9)$ is:

fully factorised

not fully factorised

wrong

prime

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	ЎЗБЕКЧА	РУССКИЙ
Method	Usul	Способ
Common factor	Umumiy ko'paytuvchi	Общий множитель
Abridged formula	Qisqa ko'paytirish formulasi	Формула сокращённого умножения
Grouping	Guruhlash	Группировка
Trinomial	Uch had	Трёхчлен
To split a term	Hadni ajratish	Разбить член
Completely factorised	To'liq ajratilgan	Полностью разложен
Order of steps	Qadamlar tartibi	Порядок действий

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The first method to try is always:

a formula

the common factor

grouping

splitting

Grouping is used when there are:

two terms

three terms

four terms

one term

For $x^2 + px + q$ the two numbers have product:

p

q

$p + q$

pq

$x^2 + x - 12$ factorises to:

$x^3 - x$ factorises to:

A factorisation is checked by:

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $x^2 + 7x + 12$

ANSWER $(x + 3)(x + 4)$

2. $x^2 - 5x + 6$

ANSWER $(x - 2)(x - 3)$

3. $x^2 + x - 12$

ANSWER $(x + 4)(x - 3)$

4. $x^2 - 2x - 15$

ANSWER $(x - 5)(x + 3)$

5. $x^2 - 9$

ANSWER $(x - 3)(x + 3)$

6. $6x^2 + 9x$

ANSWER $3x(2x + 3)$

7. $x^2 + 6x + 9$

ANSWER $(x + 3)^2$ **Medium · the standard question**

7 PROBLEMS

1. $2x^2 - 18$

ANSWER $2(x - 3)(x + 3)$

2. $3x^2 + 12x + 12$

ANSWER $3(x + 2)^2$

3. $x^3 - x$

ANSWER $x(x - 1)(x + 1)$

4. $2x^2 + 10x + 12$

ANSWER $2(x + 2)(x + 3)$

5. $x^2 - 8x + 15$

ANSWER $(x - 3)(x - 5)$

6. $x^2 + 2x - 8$

ANSWER $(x + 4)(x - 2)$

7. $ax + ay + bx + by$

ANSWER $(x + y)(a + b)$

Hard · stretch and reason

7 PROBLEMS

1. $a^3 + a^2 - a - 1$

ANSWER $(a - 1)(a + 1)^2$

2. $x^4 - 5x^2 + 4$

ANSWER $(x - 1)(x + 1)(x - 2)(x + 2)$

3. $3x^3 - 27x$

ANSWER $3x(x - 3)(x + 3)$

4. $x^2 + x + 1$

ANSWER Prime — it does not factorise

5. $2x^3 + 16$

ANSWER $2(x + 2)(x^2 - 2x + 4)$

6. $x^2 - 10x + 25 - y^2$

ANSWER $(x - 5 - y)(x - 5 + y)$

7. $4x^2 - 12x + 9$

ANSWER $(2x - 3)^2$

07 Homework**Homework — 5 tasks**

Take the common factor out first, and expand your answer back at the end.

- Factorise $x^2 + 9x + 20$ and $x^2 - 7x + 10$.
- Factorise $x^2 - 3x - 10$.
- Factorise $5x^2 - 45$ completely.
- Factorise $x^3 + 2x^2 - x - 2$ by grouping.
- Say which method each of $4a^2 - 9$, $6ab + 9a$ and $x^2 - x - 6$ needs.